

Semantics Delivery Network: Rethinking Web Retrieval Infrastructure for LLM Agents

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Abstract

Large language models (LLMs) increasingly rely on external sources when answering questions that require proprietary information or up-to-date live web content, through both traditional single-shot retrieval-augmented generation (RAG) and multi-turn agentic RAG. Yet today’s web infrastructure is still built for human clients. Given a query, current search services return a list of URLs and snippets ranked for generic relevance; content delivery networks (CDNs) cache URL-addressed objects (texts, images, videos, etc.) without knowing which passage an agent needs. LLMs, in contrast, consume short, semantically coherent passages—hereafter, *chunks*—selected for downstream task utility rather than similarity alone, and may retrieve statefully across reasoning turns. Uncoordinated agents also repeat search, data acquisition, and semantic processing, duplicating work that could be shared. We argue that *semantic chunk retrieval should become a first-class network-delivery abstraction*. We propose *Semantics Delivery Network* (SemDN): an origin-authorized, hierarchical edge substrate that indexes, searches, and smart-caches web content at chunk granularity. SemDN serves agents on behalf of participating websites, amortizes data acquisition and processing across agents, and supports tenant-specific retrieval policies. Because, unlike URL caching, semantic retrieval provides no explicit miss signal, SemDN must estimate when its enrolled corpus may be incomplete or stale and trigger scoped discovery or refresh. It raises open questions about shareable retrieval state, hierarchical caching, coverage risk, and deployment. Our preliminary probes reveal a large gap between page content processed and chunks consumed, substantial task-local reuse, and higher answer quality per context token from chunk delivery.

CCS Concepts

• **Networks** → **Network architectures**; *In-network processing*; *Overlay and other logical network structures*; **Network measurement**; • **Information systems** → **Web searching and information discovery**.

Keywords

Content Delivery Networks, Retrieval-Augmented Generation, Agentic Search, Semantic Caching, Vector Search

1 Introduction

Machine-generated retrieval is becoming an important part of the web’s read path. LLMs retrieve external content to ground answers, from single-shot RAG [60] to multi-turn *agentic search* that interleaves reasoning and retrieval [47, 68]. Increasingly, this retrieval targets the *live* web: browsing agents and freshness-sensitive benchmarks make static corpora insufficient [58, 64, 75, 104, 109, 111], while query-time fetcher traffic is rising sharply [26, 32, 123].

Yet these machine clients retrieve through infrastructure designed for human readers. Discovery APIs expose URL- and snippet-oriented results optimized for generic relevance; CDNs cache URL-addressed objects without knowing which passage an agent needs. An LLM instead consumes short semantic chunks¹ rather than pages [84, 122]. It cares about downstream *utility*, not similarity alone [16, 28, 68, 92]; retrieves statefully across related sub-queries; and belongs to an uncoordinated population that repeatedly downloads, cleans, and embeds the same content, pushing redundant work onto origins [32].

Today, this path is split across two systems (Fig. 1a). A search engine serves the client and decides *what* to read, but it returns only URLs. A CDN serves the origin with edge placement, offload, freshness management [2], and even origin logic at the edge [24, 63], but it never sees the query.

We argue that *semantic chunk retrieval should become a first-class network-delivery abstraction*. We propose *Semantics Delivery Network* (SemDN): a new semantic delivery substrate that fuses both roles: like a search engine, it answers clients’ queries; like a CDN, it delivers origin content from the edge. Despite its name, SemDN is not a CDN add-on but is built around queries: it encodes, indexes, searches, and caches web content at chunk granularity, returning the

¹A *chunk* broadly denotes a semantic unit; this may be a text passage [53], an image region [33, 85] or page screenshot [107], a table block [40], an audio clip [112], or a video segment [71].

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passages an LLM needs rather than pages it must download and post-process.

Two recent results make this new paradigm increasingly practical. First, recompute-based indexing, such as EdgeRAG and LEANN, shows that embeddings need not all be stored but can be regenerated on demand during search [89, 108]. An edge can therefore cache versioned normalized content blocks, maintain a canonical candidate index, and selectively materialize registered model-specific vectors where demand justifies them, trading storage for compute. Second, agent query streams can be bursty and task-coherent, creating locality that an edge hierarchy may exploit for placement and reuse [29, 59].

Prior systems separately expose retrieved chunks as a service [1], build managed indexes over tenant corpora [23], tier retrieval across edge and cloud for one deployment [61], or reduce vector-index storage through on-demand recomputation [89, 108]; LLM-serving caches store responses or KV states per application [8, 48]. Concurrent visions have origins expose chunks behind a DNS-like semantic resolver [12] or distribute KV caches like CDN content [21, 86], and a major CDN now builds opt-in per-site AI indexes [22]. SemDN instead delivers content-level evidence ranked across origins, for clients as well as origins: it joins semantic selection with an origin-authorized hierarchical delivery and freshness path, sharing acquisition and normalized content across tenants while exposing semantic demand to placement and refresh decisions.

This paper makes three contributions. (1) We characterize today’s LLM web-retrieval path and distill four structural mismatches between that workload and the search-engine/CDN stack, including its redundant, failure-prone preprocessing (§3). (2) We propose Semantics Delivery Network (SemDN), a new edge retrieval substrate, and contrast its data path with today’s (§4, Fig. 1). (3) We organize its research agenda around shareable retrieval state, hierarchical caching, and two-sided, verifiable deployment, and we report preliminary evidence for semantic retrieval as a network-delivery abstraction (§5, §6).

2 Background

2.1 Content Delivery Networks

We take CDN basics as given and recall only what bears on our argument. A CDN caches origin content at edge points of presence (PoPs), keyed by URL and expired by a time-to-live (TTL) [2]. Its value is two-sided in effect but one-sided in billing: clients enjoy lower latency, while content owners—who pay the CDN—offload origin traffic, absorb flash crowds, and gain DoS protection. Two properties matter here. First, a CDN treats content as opaque bytes and places or evicts objects by popularity and recency, not by

semantic relationships. Second, although CDNs increasingly run origin computation and defense [24, 63, 113, 114], their placement remains URL-object-based.

2.2 Retrieval for LLMs

RAG augments an LLM by retrieving external passages from proprietary sources or the live web, and placing them in the model’s context [60]. A corpus is segmented into *chunks* (typically a few hundred tokens [84, 122]), each mapped to a dense vector by an *encoder*, e.g., E5 [106] or BGE [17]. A query is embedded by (typically) the same encoder, and the approximate nearest-neighbor search (ANNS) system returns the top-*k* chunks efficiently with a graph index such as HNSW [69, 72], a clustering-based index such as IVF and IMI [9, 46], or low-bit representation [36, 43, 44, 120, 121]. Embeddings need not be stored: rather than keeping an index several times larger than the raw text, a system can recompute embeddings on demand and trade storage for compute [108], a trade we return to as a design knob (§5.1).

Agentic search interleaves reasoning and retrieval over multiple turns: the agent identifies missing information, queries, incorporates returned passages, and repeats, turning one question into a task-coherent sequence [5, 47, 111]. Yet similar passages can contain no useful information for the answer and might instead mislead reasoning with distractors [27], so retrievers trained on downstream correctness rather than query–passage similarity can improve accuracy and reduce search turns. Moreover, the best retriever may be coupled with the agent [68, 92].

2.3 How LLMs Retrieve From Web Today

A common production path for LLM web retrieval is a two-stage pipeline (Fig. 1a). A search engine results page (SERP) API (e.g., Serper [90], Brave [14], Exa [31]) maps a query to ranked URLs and snippets but does not return content. A separate reader (e.g., Firecrawl [34], Jina [50, 105], Tavily [97]) fetches those pages, strips boilerplate, and returns clean text. The application then chunks, embeds, ranks, reasons, and may issue another query [47, 68, 94]. Applications cannot generally rely on cross-provider reuse or origin-integrated freshness; each assembles its own pipeline or adopts a provider-specific one.

3 Observations

We distill the gap between this workload and its infrastructure into four observations.

O1: Objective mismatch: similarity is not utility. Mainstream discovery services nowadays [14, 90] return results ranked by generic relevance (such as lexical similarity), while off-the-shelf encoders [88] optimize query–passage *similarity*; agents instead need passages that are *useful* for producing a

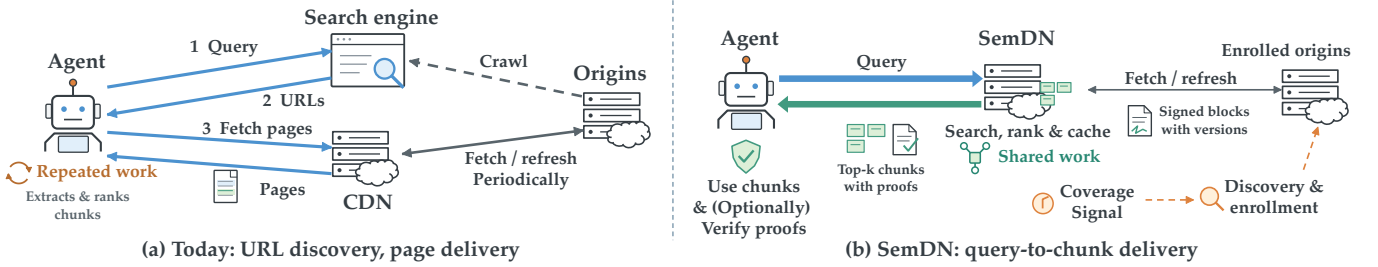


Figure 1: Today’s retrieval path versus SemDN. (a) A search engine returns URLs; the agent fetches whole pages through a semantics-agnostic CDN and extracts chunks itself. (b) SemDN answers the query with top- k chunks from a shared index over enrolled origins, optionally with proofs; a coverage signal triggers discovery and enrollment.

correct answer. The two are not always the same: similarity-trained retrievers underperform utility-aware ones [68, 126], and highly similar passages can even derail reasoning [27]. Search-R1 uses answer-level RL to adapt a search policy to a fixed retriever; Agentic-R further trains the retriever on local relevance and global answer correctness [47, 68]. This trend motivates tenant-specific utility ranking rather than treating shared similarity as the final objective. Content acquisition and canonical candidates can be shared; a tenant’s utility function and outcome labels need not be. Our preliminary results provide motivating evidence: similarity-ranked page-first retrieval leaves useful chunks unreached and is matched or outperformed by chunk-first retrieval (§6.3); whether a utility-customized chunking and retrieval pipeline widens this gap is future work.

O2: Granularity mismatch—pages versus chunks. The web is authored, crawled, and delivered at the granularity of pages and their resource-heavy objects, whereas LLMs consume shorter passages or chunks [1, 84, 125]. CDNs cache URL-addressed objects, not passages. An HTML-only fetch still transfers the full document; browser-rendering agents may additionally retrieve its subresources—the median desktop page is now ≈ 3 MB across roughly 70–77 requests [42], while the passage a model wants is typically a few kilobytes. Today’s agents bridge this gap on the client side expensively: many production agents render complete pages and operate over *screenshots* or other browser-level representations [4, 37, 54, 78, 79], fetching and processing essentially the whole page to read a kilobyte-scale slice. The useful passage is thus one to three orders of magnitude smaller than what is fetched, leaving large room for optimization.

Furthermore, chunking is a *budget* decision in addition to a bandwidth one: stuffing whole pages into the model’s context inflates inference costs and potentially degrades answer quality [67]. Therefore, how content is chunked and selected is part of answering well—and it is itself non-trivial and improvable (fixed-size windows, neural text segmentation [70, 84], etc.), and LLM-driven chunkers that split by semantic logical

boundaries or into self-contained propositions [19, 20, 30, 125]. The best granularity varies per query [119, 127]. Existing mitigations optimize this work on the client [13, 20, 66, 96] or convert whole pages to Markdown at the CDN [25]; we ask whether a shared edge substrate can instead perform it once, well, and amortize it (§4).

O3: Locality and policy mismatch—URL versus semantics. CDN caching keys on URL and recency; LLM retrieval keys on semantics and can exhibit locality that the current stack cannot see. AI scrapers and search agents already degrade URL caches [123]: even a search agent draws 70–100% unseen URLs within a task [123], yet in our traces 32–55% of a new query’s chunks recur within the task (§6.2). Reuse invisible to URLs reappears at chunk granularity. Query streams in RAG workloads are bursty and task-clustered [29, 59, 111]: within one task and across similar tasks, retrieved chunks recur across turns. Separating discovery from URL-object delivery prevents the delivery layer from directly exploiting this semantic locality for cross-client placement and eviction.

O4: Redundant, failure-prone semantic preprocessing. Turning a URL into model-ready chunks is a heavy, multi-stage pipeline that requires reaching the page (often past bot walls [80], JavaScript shells, and timeouts), extracting content from a noisy web, stripping boilerplate, segmenting, and embedding [20, 35, 95, 96, 105]. Because each application assembles this pipeline itself (§2.3), popular pages may be processed independently by many operators. The work is both substantial and unreliable: across 1,430 live SERP result URLs, a plain HTTP client from our vantage point obtains usable clean text from only 37%, and even a cloud renderer from at most 75%, after which the clean text must still be de-boilerplated (raw extraction carries $\approx 2\times$ the needed text) and embedded. This fetch-clean-chunk-embed work can be repeated across tenants within a single freshness window while origins absorb the resulting fetch storms [32]. Acquisition and normalization can instead run once per content version, while representation-specific embedding and indexing run once per registered retrieval configuration (§6.2).

4 Semantics Delivery Network (SemDN)

4.1 Architecture

SemDN introduces a semantics-aware delivery path between agents and participating origins. In the SemDN data path (Fig. 1b), a client (e.g., a RAG application or an agent) sends a *query*, not a URL, to its nearest edge. We assume a mature SemDN already maintains a large-scale, origin-authorized semantic corpus, populated by participating origins and prior acquisition, rather than constructing that corpus from a single query. The edge runs three core functions: encode (map queries and content candidates to vectors), search (over canonical or registered model-specific indexes), and cache (store and evict normalized blocks, materialized chunks, and index state). Borrowing the hierarchy that makes CDNs scalable, an edge PoP holds hot content and index state, regional parents hold larger, colder shards, and a global tier maintains the distributed catalog; resource selection can avoid searching every shard [3].

This hierarchy changes the meaning of a cache miss. A URL cache, like an information-centric network that routes on exact content names [45], has a binary exact-match miss; semantic search replaces names with ranked queries and always returns top- k results, even when newer or better evidence lies outside its index. We define a *semantic coverage miss* to occur when the estimated risk that external discovery would add fresh evidence and change the returned top- k exceeds a policy threshold. The coverage controller could combine retrieval-score distributions and result coherence [93, 100, 101, 103], judgments of whether the retrieved evidence suffices to answer [51], cross-tier agreement, and freshness metadata as imperfect signals of coverage risk. A judgment reading only query and evidence, not the tenant’s generator, can run once in the network before generation. Periodically, or on a high-risk query, it invokes an external discovery service (e.g., a SERP API), as Corrective RAG does for low-confidence retrieval [115]; results are filtered against the origin-enrollment registry before SemDN fetches or revalidates absent or stale content. FreshCache gates per-client semantic-cache reuse on an estimated staleness probability [73]; SemDN applies a similar risk test to a shared, origin-authorized corpus. Calibrating this risk under workload and content drift is an open problem.

4.2 Design Decisions

Four decisions follow from the observations and motivate the agenda in §5.

Share normalized content and canonical candidates; specialize ranking per tenant (O1). Different chunkers and encoders induce different objects and proximity graphs, and

single-vector embeddings cannot express every top- k ranking [110]. SemDN does not assume that one index serves arbitrary tenant policies. It caches versioned normalized content blocks—e.g., DOM paragraphs, table blocks, image regions, and provenance—from which supported chunking policies compose passages. A shared canonical index retrieves candidates; a tenant-provided utility reranker or a pre-registered model-specific encoder refines them, with specialized index state materialized only where demand justifies it. Tenant utility functions and outcome labels remain private (§5.1).

Serve and cache at chunk granularity (O2). The edge returns the passages a model consumes, not the pages it would otherwise download and post-process. Normalized blocks remain the stable shared objects; hot chunks composed under supported policies can be materialized and cached at the granularity the workload actually reuses (§6.1, §6.2).

Place and evict in the query stream (O3). Because SemDN sees the agent query stream, it can place and evict based on observed semantic reuse rather than URL popularity alone. How much query state to share across tenants is open (§5).

Do shared work once and specialized work once per registered representation (O4). Acquisition, cleaning, and normalization run once per content version; canonical embedding and indexing are shared across tenants, while registered model-specific state is amortized among tenants that use it. This relieves origins of redundant fetches without pretending that arbitrary embedding spaces share one index (§6.2).

5 Architectural Research Agenda

SemDN raises three coupled architectural questions: what normalized content and retrieval state is **shareable** across heterogeneous tenants; how to make a hierarchical substrate **manageable** for the discovery, caching, and refreshing of web content under semantic coverage misses; and what tenant interfaces, origin controls, incentives, and verifiability guarantees make the resulting service **deployable**.

5.1 What Retrieval State Can Tenants Share?

Representation. What stable *form* should the cached asset take? SemDN can store the versioned normalized blocks of §4, then compose tenant-visible chunks through supported policies. A canonical embedding or a Matryoshka representation, truncatable to smaller dimensions [55, 87], enables a shared first-stage candidate index but presumes an agreed encoder. A subtlety often missed: even with recompute, the proximity graph is *metric-specific*—an HNSW graph encodes nearest-neighbor relations in one embedding space and does not transfer to another [72, 108]. Hence, a model-specific graph is shared only among tenants registered for the same representation.

Indexing and the storage–compute trade. SemDN’s baseline uses a canonical index for candidate generation; sufficiently popular registered encoders may add model-specific indexes. For either option, storing embeddings versus recomputing them remains flexible. LEANN [108] and MHR [43] can reduce the storage requirement of graph-based indexes through lightweight structures [46], at the expense of additional GPU load, whose viability depends on edge compute headroom. Where a real edge should sit on this surface, and whether PoPs carry the assumed compute, is left open.

5.2 How Should a Hierarchical SemDN Cache and Refresh?

Caching, locality, and coverage. This is the richest open area (O3). Where AI-aware web caches filter training scrapers and search agents alike as low-locality load [123], SemDN serves query-time retrieval, not training scrapes: an agentic task, or an organization’s agents, forms a cohort whose reuse follows semantics. Agent trajectories reveal which chunks are reused together [59], while cross-tenant demand determines which content merits edge placement rather than a regional-only copy. Approximate reuse of retrieval results across similar queries [11, 39, 77] could extend hits beyond exact repeats.

The challenges include (1) *coverage skew* (only what is queried gets optimized, leaving the long tail under-served); (2) *cross-tenant leakage* (reweighting a shared structure from one tenant’s trajectories can expose its task structure to others); and (3) *feedback entrenchment* (favoring previously successful chunks risks locking in stale answers). The controller must also distinguish an edge miss, which a regional parent can satisfy, from a system-wide semantic coverage miss that warrants external discovery: false confidence hides fresh evidence, and outdated evidence retrieved alongside current evidence actively degrades answers [81], while a conservative threshold restores redundant search and origin load. Sufficient evidence may also be superseded, so coverage risk must weigh freshness too.

Freshness. SemDN can adapt CDN-style freshness machinery to amortize *acquisition*. For example, a query that forces a re-fetch can refresh related chunks (spatial locality), and content warmed by one tenant’s traffic serves the cohort. Versioned blocks also suit evolving documents, where amendments change few clauses but retrieval must select the right version [76]. This advantage is not unconditional. Freshness-critical content (news, prices) limits cache lifetimes, periodic TTL refresh provably leaves stale windows [62], graph indexes must absorb frequent in-place updates [65], and cold content may never be warmed by piggybacking. So we treat the achievable savings as something to estimate rather than assume.

5.3 What Interfaces, Incentives, and Guarantees Make SemDN Deployable?

Tenant interface. Today’s agents retrieve from the web through SERP and reader APIs (§3) or managed indexes [23]. They return ranked snippets, cleaned text, or a synthesized answer along with a bill, but generally expose limited control over *why* a passage was returned or how retrieval policy affects it. Yet the embedding model and chunking strategy materially change which passages are retrieved and how well the agent answers [15, 74], so a fixed, hidden choice can cap quality, and a per-query bill prices a process the client cannot inspect.

SemDN instead lets a client select a supported chunking policy over normalized blocks and either use canonical retrieval with a tenant utility reranker or register a model-specific encoder (§5.1). It returns selected passages with scores and provenance, making retrieval auditable and billable for what was consumed. Managed services already offer website ingestion, configurable chunking/models, reranking, citations, semantic query caching, and tenant isolation [23]; CDN operators add opt-in per-site AI indexes with change feeds [22]. Both are organized around the origin: one index and representation per site, federated by fan-out. SemDN instead ranks chunks across origins (§6.3) and shares origin-authorized acquisition and normalized content across independent retrieval clients, coordinating placement, freshness, and representation-specific state around client demand. For clients, this transparency is as valuable as the content volume it saves.

A two-sided service. A CDN is a reverse proxy paid by the origin; an enterprise web proxy is a forward proxy for its clients. **SemDN is both and can monetize two services.** To clients, including organizations whose agents share locality, it sells *retrieval quality and latency*: instead of fetching full pages and running the whole pipeline, RAG systems receive model-ready chunks, billed per chunk consumed and priced along the representation/accuracy surface (§5.1).

To content owners, it sells origin offload for machine traffic, analogous to a CDN: SemDN terminates agent fetches at the edge and serves only the relevant chunks rather than full pages, reducing both origin load and delivered bytes while retaining freshness management and DoS defense. An origin contract defines enrollment, permitted transformations, freshness/version bounds, purge and invalidation, access control (agent traffic is already identifiable at the TLS/HTTP layer [52]), and provenance; the interface could expose aggregate machine-demand statistics while withholding individual queries. Pricing can also steer architecture: charging registered model-specific representations at their unshared cost while discounting a canonical representation gives tenants an incentive to use shareable state. Settlement is the main

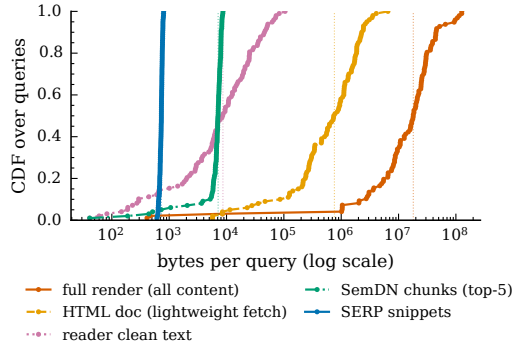


Figure 2: CDF of decompressed content bytes per query over the top-5 URLs (live web, 100 queries); values in the text are medians.

adoption hurdle: neither side pays merely because the other uses SemDN.

Verifiability. Acting for both sides creates conflicts a CDN has never faced: an operator paid by origins may rank paying origins higher, and a shared cache can leak tenants’ queries through timing, as in shared prompt caches [38]. HTTPS authenticates only the endpoint and byte stream: a CDN already terminates TLS with the origin’s delegated credentials [10], and a chunk is byte-identical to nothing the origin published, so a faulty edge can cherry-pick, suppress, or inject passages [128]. Clients must instead verify the *evidence-construction process*: that blocks come from their origin at a stated version (e.g., via HTTP message signatures [7]); that a chunk is composed of those blocks (via Merkle inclusion proofs); that the top- k is the honest result over a committed index, which also exposes pay-to-rank bias [6, 18, 83]; and that the index omits no enrolled origin, as transparency logs ensure for certificates [57]. Verifiable HNSW search takes 0.8–2 s per query [6]; amortizing audit-on-demand proofs [83] across cohorts and tiers is an open networking problem.

6 Preliminary Results

We report preliminary evidence along three axes. Two concern *cost*: the content volume an agent processes to obtain the text it consumes (§6.1; O2, O4), and the locality and cross-tenant amortization a chunk cache enables where a page cache cannot (§6.2; O3–O4). The third concerns *quality*: chunk-granular delivery matches or beats the page and search-engine baselines in most settings, with far less token budget and therefore lower inference cost (§6.3; O1–O2). Throughout, a *chunk* is a passage of ≈ 100 –256 tokens and a *page* is its source web page as a whole; an encoder maps text to vectors for top- k ANN search and a *reranker* re-scores candidates.

6.1 The Page Tax: Bytes Processed vs. Used

We send 100 real queries (50 curated intents across ten categories plus 50 from Natural Questions [56]) through the production agent path of Fig. 1a: a SERP API returns the top-5 URLs, which we then fetch and process. Across the acquisition spectrum, five sharply separated byte bands emerge—SERP snippets, SemDN chunks, reader clean text, HTML documents, and full renders (Fig. 2). At the median, the model consumes **7.1 KB** of selected chunks, while the lightweight no-JS baseline processes **761 KB** of decompressed HTML to obtain them ($107\times$); a full browser render yields **18.1 MB** of decompressed content ($2,532\times$, or 5.75 MB on the wire). Plotted values are post-decompression processing volumes; wire values are labeled separately. Fetching is also brittle: at least 39% of the 500 sampled URLs lose over half their text without JavaScript and $\approx 19\%$ are bot-blocked, making fetch-clean-chunk-embed heavy and failure-prone when performed independently by each client (O4).

6.2 Agentic Locality

Observed session locality. We run Search-R1 on Qwen2.5-7B [47, 116] over wiki-18 [49, 53] (21M chunks, 3.2M pages) with the fixed e5-base-v2 backend to which its search policy was adapted during RL [106] and log every retrieval in a multi-session stream. The locality O3 predicts is large and *task-driven*: within a task, 30.5% of post-first-turn queries are byte-identical repeats, consistent with the redundant retrieval reported for Search-R1 [91], and even after deleting every repeat, 32–55% of a new query’s chunks were already retrieved earlier in the task. Compared with popularity-preserving, task-shuffled controls, the real stream reuses chunks after 1–2 orders of magnitude fewer intervening chunks—structure a URL/TTL cache cannot key on (Fig. 3a).

A session cache captures exact repeats; SemDN additionally enables cross-client reuse, shared acquisition and freshness, and origin offload.

Trace-driven cache-byte savings. We replay the stream against simulated PoP caches. In the evaluated URL-object cache, an entry is a source article sized as the sum of its chunk bytes, and a miss fetches the entire article. This favors the URL-object cache by excluding markup and subresources. A chunk-granularity LRU cache of just **1.34 MB** serves **73.8%** of chunk requests (87.6% at 13.4 MB). The URL-object LRU under this page-first delivery path reaches a comparable hit ratio only near 100 MB and fetches **6.9–15.5 $\times$** more bytes than the agent consumes; at equal cache capacity, the chunk cache sends **13–41 $\times$** fewer bytes to origin (Fig. 3b). The study replicates with the same agent over a realistic web corpus from CRAG, where within-task reuse is stronger (63–79%) and the chunk cache sends 25–70 $\times$ fewer origin bytes at equal size.

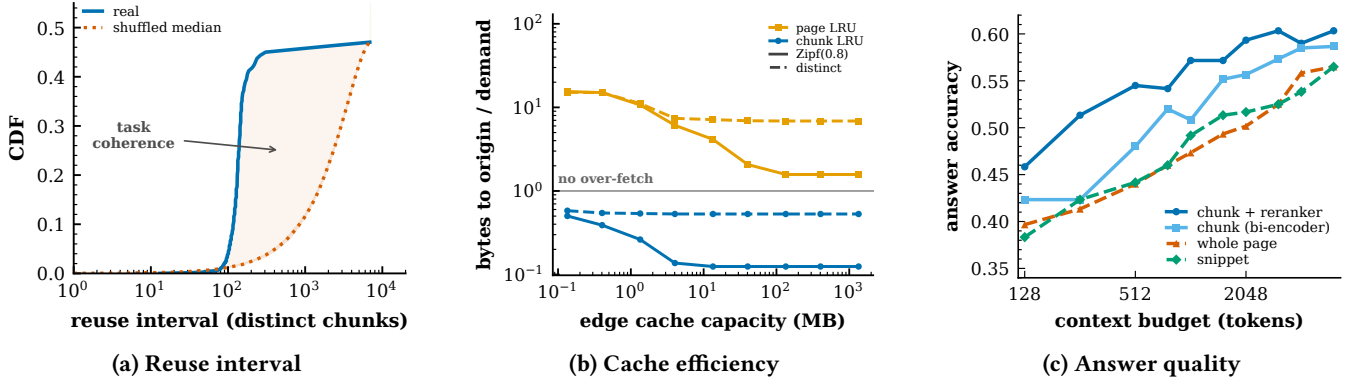


Figure 3: Agentic locality (a,b) and answer quality (c). (a) Reuse-interval CDF for distinct chunks; shaded band shows 5–95% over 50 popularity-preserving, task-shuffled controls. (b) Bytes to origin per demand byte vs. edge cache capacity: a chunk cache sends far less to origin than a page cache at every size. (c) Answer quality vs. reader context budget (CRAG Task-3, $n=600$; Qwen3-32B reader, Llama-3.1-70B judge [98]): reranked chunks answer more per token than pages or snippets at every budget.

Modeled cross-tenant amortization. The top-10 domains draw 34% of results across our live set, suggesting potential overlap but not establishing that tenants request the same pages or chunks. We therefore feed the measured per-page and embedding costs (392 chunks/s per A100 GPU with e5-base-v2) into a Zipfian tenant-demand model, which estimates $\approx 6\times$ less repeated fetch-clean-chunk-embed work at 50 tenants. Embedding and index work amortize only among tenants sharing the canonical or same registered representation. A private per-tenant encoder remains $1\times$ in the model, although acquisition and normalized blocks remain shareable—the incentive for encoder-aware pricing in §5.3.

6.3 Answer Quality

In the evaluated fixed-corpus pipeline, chunk-first retrieval performs better than page-first retrieval. Using e5 over wiki-18 with five QA sets—HotpotQA [118], 2WikiMultiHopQA [41], MuSiQue [102], Bamboogle [82], and NQ [56]—and the same Qwen2.5-7B reader, chunk retrieval with the *small*, fixed e5 backend used by Search-R1 matches or beats page retrieval with a *strong* 32k-context embedder (Qwen3-Embedding [124]) on four of the five, improving average F1 from 0.334 to 0.357 while delivering $\approx 470\times$ fewer bytes. On 300 time-stable HotpotQA questions, 16–29% of SemDN’s global top-5 chunks lie on pages that a whole-page ranker does not select, even at 20 downloaded pages, which source-first designs would miss [12]. A client that instead ranks pages by their best chunk needs five pages (0.76 MB, $\approx 238\times$ the bytes) to recover the same chunks; read by the same model, SemDN’s five chunks (3.2 KB) beat the page-first client’s F1 even at *ten* downloaded pages (1.5 MB, $\approx 470\times$).

We also test a noisier, web-derived benchmark regime. On CRAG [117], we answer each question once with a Qwen3-32B reader [99] and vary only how the retrieved content is packed into the reader’s context. Under equal token budgets over the same cleaned content, relevance-ranked chunks match or outperform whole pages and SERP snippets across the 128–6,144-token sweep (Fig. 3c). Cross-encoder reranking reaches 0.60 accuracy, with the largest gains at tight budgets. This is consistent with the risks of stuffing long pages into context [67].

7 Conclusion

LLM agents consume task-relevant passages, yet today’s web stack discovers and delivers URL-addressed objects. We argue that semantic selection should become a first-class network-delivery abstraction. Our probes show that page-first retrieval processes far more content than agents ultimately consume, that agent queries exhibit reusable locality, and that shifting toward chunk delivery improves quality per context token, hinting at significant opportunities for such a semantics-aware network substrate. Open challenges in coverage, indexing, freshness, privacy, incentives, and verifiability define a new networking research agenda.

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